Nome _____ Cognome _____ Data ____/____/_____

Thermal physics: test

Version A

Time: 50 minutes. Total: 30 marks. Calculator, ruler and sharp pencil allowed. Equations are not given; data and constants are at the end of Part B. In Part A and in Part C circle one letter for each question, except question 21 (write the answer on the line). In Part B, questions 13 to 16, write only the final answer on the line; in questions 17 and 18 write the three lines asked for (there is some space for working if you want it).

A Multiple choice [12]

Choose one letter for each question. Each question is worth 1 mark.

1. Two rods, X and Y, have the same length at 20 °C. Both are heated by 50 °C. Rod X becomes 1.8 mm longer and rod Y becomes 1.2 mm longer. What does this show? **[1]**

A X has a larger specific heat capacity than Y.

B X has a larger linear expansion coefficient than Y.

C X absorbed more energy than Y.

D X was longer than Y before heating.

2. Two beakers contain water at 60 °C. Beaker P holds 100 g of water and beaker Q holds 300 g. Which statement about the water particles is correct? **[1]**

A The particles in Q have more average kinetic energy, because there is more water and so more energy to share.

B The particles in P have more average kinetic energy, because the same energy is shared among fewer particles.

C The particles in Q move faster, because a larger mass of water needs more energy to reach 60 °C.

D The particles in P and in Q have the same average kinetic energy, because the temperature is the same.

3. A fixed mass of gas is sealed in a rigid container and heated. The pressure of the gas rises. Why? **[1]**

A The particles move faster, so they hit the walls more often and with more force.

B The particles become larger and take up more space.

C The number of particles in the container increases.

D The particles are pushed closer together by the heating.

4. Why does steam at 100 °C cause a worse burn than the same mass of boiling water at 100 °C? **[1]**

A Steam particles have a larger mass than water particles.

B Steam is at a higher temperature than boiling water.

C When the steam condenses it releases a large amount of energy (latent heat) onto the skin.

D Steam has a larger density, so it presses harder on the skin.

5. Why does the temperature of a liquid fall as it evaporates? **[1]**

A The particles that escape take their energy with them, but they are chosen at random, so the average energy of the particles left behind does not change.

B The particles with the most energy are the ones that escape, so the average energy of those left behind is lower.

C The liquid loses particles, so there is less liquid to heat and its temperature falls.

D Energy is absorbed from the surroundings to make the particles escape, so the surroundings cool down while the liquid stays at the same temperature.

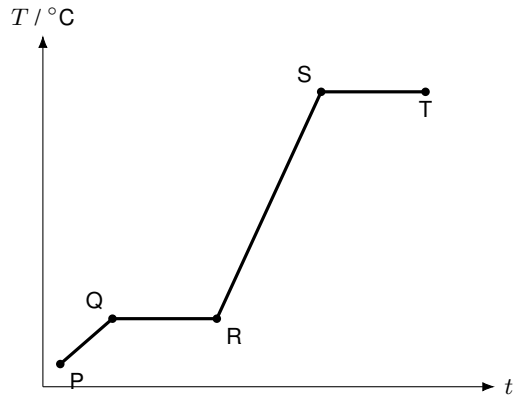


Fig. A (not to scale)

6. A solid is heated at a constant rate until it has become a gas. Fig. A shows how its temperature changes. In which section is the substance entirely liquid, with its temperature rising? **[1]**

A PQ
B QR
C ST
D RS

7. A fleece jacket keeps you warm because it traps air between its fibres. Why does trapped air reduce the energy lost from the body? **[1]**

A Air is a good conductor, so it spreads the energy evenly over the body.
B Trapped air absorbs all the infrared radiation from the body.
C Trapped air is hotter than the body and warms it.
D Air is a poor conductor, and trapped air cannot circulate, so there is little convection.

8. The heating element of an electric kettle is at the bottom. Why does all the water become hot? **[1]**

A Hot water is denser than cold water, so it sinks and pushes the cold water upwards.
B Water is a good conductor, so energy passes quickly through the whole kettle.
C Hot water is less dense and rises, cold water is denser and sinks, setting up a convection current.
D Radiation from the element is absorbed by all the water at once.

9. In the cooling experiment the hot potato was wrapped in aluminium foil. Aluminium is a good conductor, so why did the foil reduce the cooling? **[1]**

A The shiny surface of the foil is a poor emitter of infrared radiation, so it reduces the energy lost by radiation.
B Aluminium is a good conductor, so it spreads the energy evenly over the potato and slows down the cooling.
C The foil absorbs the infrared radiation from the room and passes it to the potato.
D The foil stops conduction, because metals are good insulators when they are thin.

10. Pieces of ice at 0 °C are stirred into warm water in an insulated cup. After a few minutes some ice is still present. What is the temperature of the contents of the cup? **[1]**

A Halfway between 0 °C and the starting temperature of the water.
B 0 °C
C Below 0 °C.
D The starting temperature of the water.

11. In the law of cooling, $\ln(T - T_{\text{room}}) = \ln(T_0 - T_{\text{room}}) - kt$. Two cups of tea in the same room have $k_1 = 3.0 \times 10^{-4}$/s and $k_2 = 6.0 \times 10^{-4}$/s. Which statement is correct? **[1]**

A Cup 2 approaches room temperature more quickly than cup 1.
B Cup 2 ends at a higher temperature than cup 1.
C Cup 1 cools more quickly, because its value of k is smaller.
D The two cups cool at the same rate, because they are in the same room.

12. A student measures the temperature of a bath five times and takes the average. Which type of error does averaging reduce? **[1]**

A The zero error of the thermometer.
B Both random and systematic errors.
C Neither: averaging cannot change an error.
D Random error: the readings scatter a little above and below the true value.

B Short answers [14]

Questions 13 to 16 are worth 2 marks each, questions 17 and 18 are worth 3 marks each.

13. An aluminium rod is 2.0 m long at 20 °C. It is heated to 70 °C. Calculate the increase in its length. Give your answer in millimetres. **[2]**

Answer _____ mm

14. An immersion heater of power 800 W heats 1.5 kg of water from 15 °C to 55 °C. Calculate the time taken. Assume that no energy is lost. **[2]**

Answer _____ s

15. A heater of power 1000 W boils away 0.15 kg of water that is already at 100 °C. All the energy from the heater is absorbed by the water. Calculate the time taken. **[2]**

Answer _____ s

16. A 0.50 kg block of metal is heated from 20 °C to 60 °C. It absorbs 7.8 kJ of energy. Calculate the specific heat capacity of the metal. **[2]**

Answer _____ J/(kg °C)

17. 0.40 kg of water at 80 °C is mixed with 0.60 kg of water at 20 °C in an insulated container. Calculate the final temperature. **[3]**

Equation (in symbols) _____

Substitution (with numbers) _____

Answer, with its unit _____ °C

18. A copper rod is 3.0 m long. When it is heated its length increases by 1.53 mm. Calculate the rise in temperature of the rod. **[3]**

Equation (in symbols) _____

Substitution (with numbers) _____

Answer, with its unit _____ °C

Data and constants

Specific heat capacity of water	4200 J/(kg °C)
Specific latent heat of vaporisation of water	2.3×10^6 J/kg
Linear expansion coefficient of aluminium	23×10^{-6}/°C
Linear expansion coefficient of copper	17×10^{-6}/°C
Law of cooling	$T - T_{\text{room}} = (T_0 - T_{\text{room}})\, e^{-kt}$, so $\ln(T - T_{\text{room}}) = \ln(T_0 - T_{\text{room}}) - kt$

C Alternative to practical [4]

A cup of tea is left to cool in a room at $T_{\text{room}} = 20.0$ °C. A group measures its temperature T every five minutes for 20 min and plots $\ln(T - T_{\text{room}})$ against the time t in Fig. 3. The law of cooling is on the data page.

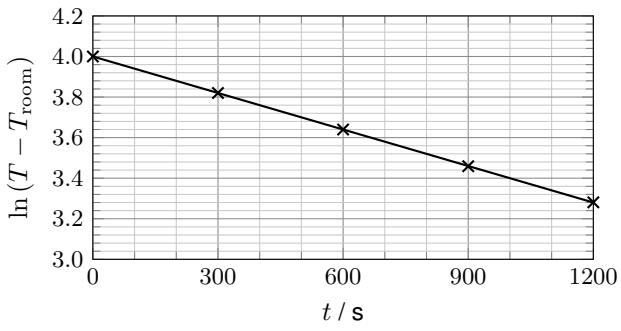


Fig. 3

19. Use the best-fit line in Fig. 3. What is the value of k in the law of cooling? **[1]**

- **A** -6.0×10^{-4}/s
- **B** 0.72/s
- **C** 1.7×10^{3} s
- **D** 6.0×10^{-4}/s

20. The tea cools from the start of the recording. Use Fig. 3 to find the time at which the tea was at 50.0 °C. **[1]**

- **A** 1000 s
- **B** 150 s
- **C** 1200 s
- **D** 1700 s

Fig. 4 shows how the temperature T of a cup of coffee, X, changes with the time t after it is poured at $t = 0$. The room is at 20 °C. A second identical cup, Y, was filled earlier with coffee at the same starting temperature and left in the same room. At $t = 0$ the temperature of cup Y is 40.0 °C.

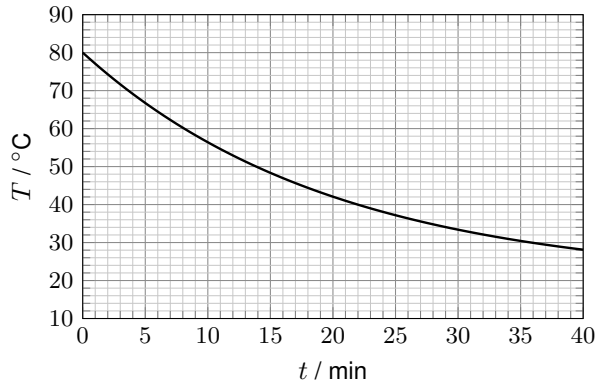


Fig. 4 cup X

21. Use Fig. 4 to find how long before $t = 0$ cup Y was filled. Give your answer in minutes. **[1]**

Answer _____ min

22. At $t = 0$, which cup is losing energy to the room faster, and why? **[1]**

- **A** Cup Y, because it has been cooling for longer.
- **B** Cup Y, because it is closer to the room temperature, so the air takes its energy more easily.
- **C** Cup X, because its temperature is further above the room temperature.
- **D** Both at the same rate, because they are identical cups in the same room.